

— TRAINING-FREE · NO TASK HEAD · OPEN VOCABULARY

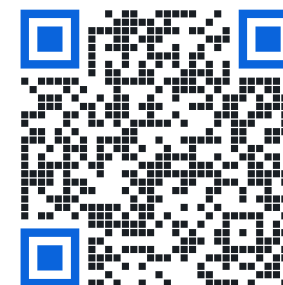
Test-time compute of jina-embeddings-v5-omni for image tagging

Han Xiao

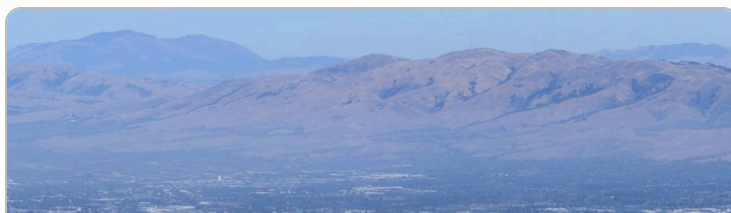
VP of AI, Elastic

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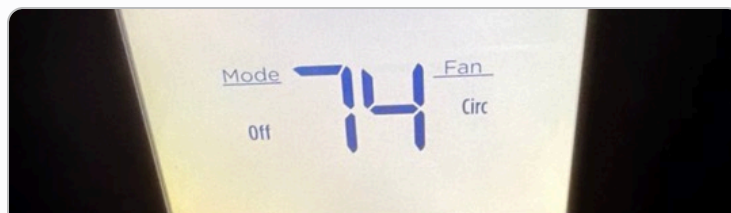
THIS DECK



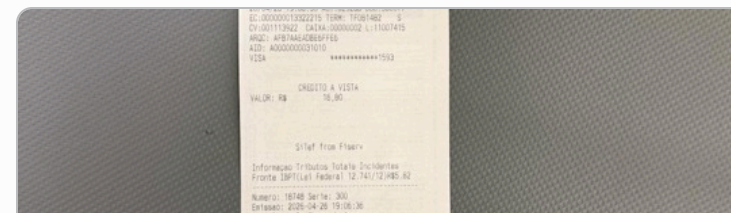
Training-free, open-vocabulary tagging from a frozen embedding model.



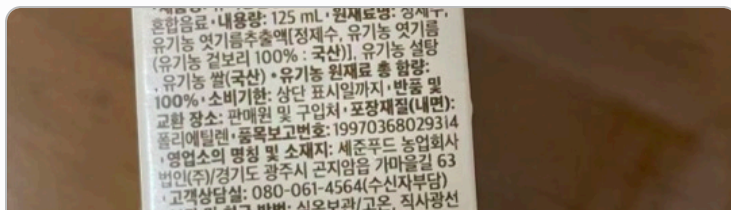
mountain · phoenix · sky · rooftop · location



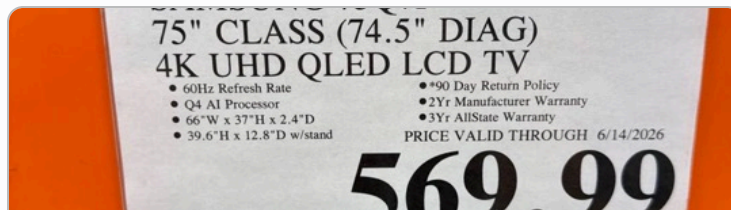
clock · thermostat · dial · dst · wifi



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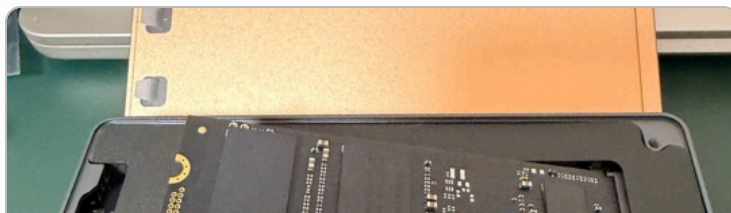
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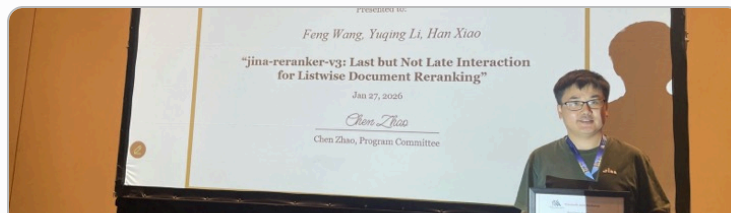
harga · lcd · samsung · refurbished · warrants



sunglasses · watches · branded · bronze · necklace



tablet · motherboard · wearable · celular · datastore



award · forwarding · speaking · defendant · workshop



gift · toy · celebration · balloons · cheerful

Problem formulation.

- | GIVEN one embedding model, **jina-embeddings-v5-omni-nano** (~1B, image and text in one 768-d space), and one image.
- | OUTPUT words for **every** object in the image: multi-label, open vocabulary.
- | FROZEN all weights. The model is a retrieval encoder; it has no tagging head, no classifier, and was never trained for this task.
- | ALLOWED inference-time computation over the model's own outputs, and nothing else.

NO TRAINING

NO SECOND MODEL

NO WORDNET / DICTIONARY

NO REGEX / POS TAGGER

Whatever tagging ability appears must come from [test-time compute of the embedding model](#).

Where new information comes from.

A frozen encoder's inference-time budget takes roughly three forms:

A DEEPER PASS

Read more of the one pass you already ran: every patch vector, not just the pooled one. The retrieval world's **late interaction**.

new information: reclaimed from inside the pass

B MORE PASSES

Run the frozen encoder again on new views: split a document, crop an image, re-encode.

new information: acquired from the input

C BOOTSTRAP

Transform the vectors you already have: whitening, propagation, optimal transport. Where most of the training-free literature lives.

new information: none

Same frozen encoder in all three; they differ only in **what enters the computation**.

Two research questions.

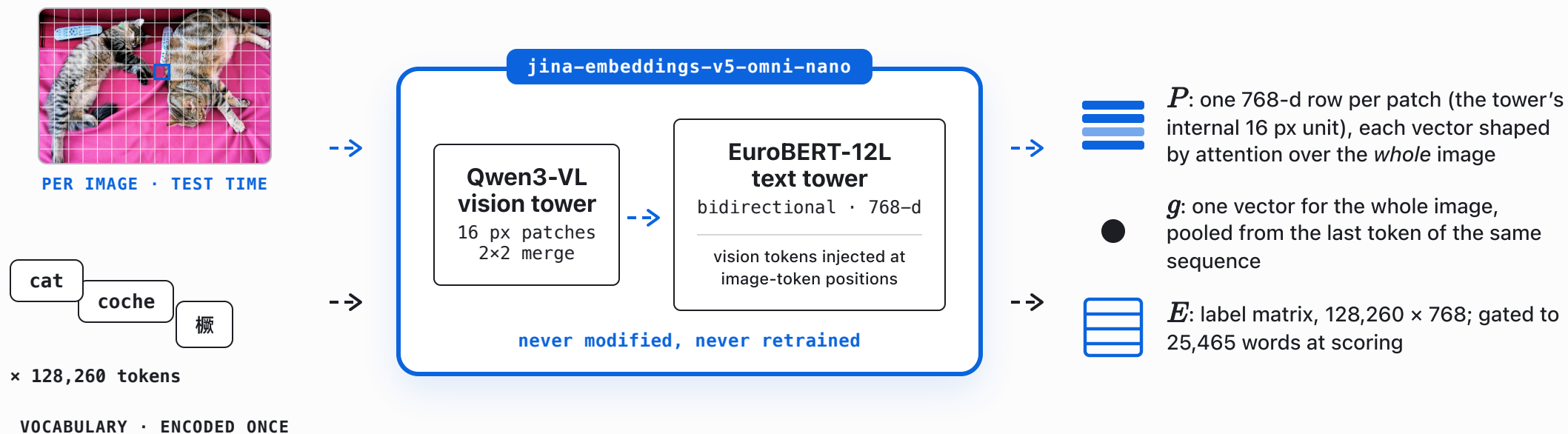
RQ1 Can a frozen embedding model become an image tagger through test-time compute alone?

Prior tagging work pays with training or resources: Tag2Text / RAM / RAM++ train over ~6,400 curated tags; TagCLIP (AAAI'24) and PIAA ("[CLS] is not enough") are training-free but assume a given class list, on two-tower CLIP plus WordNet or POS tools.

RQ2 If it can: which test-time compute scales its accuracy, bootstrapping or new information?

The 2024–26 literature bets on bootstrapping: whitening (PIAA), label propagation (ZLaP), optimal transport (OTTER), Bayesian adaptation (BCA).

Image and text embedding space.

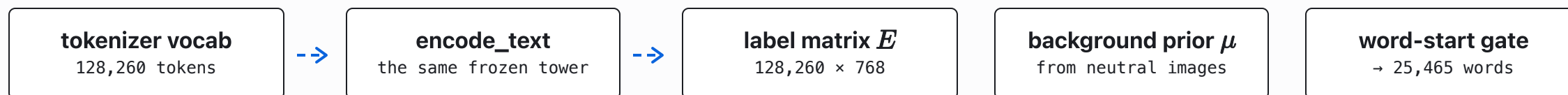


Both paths end in the same space, so $\langle p, e_\ell \rangle$ is well-defined: **any patch of any image, scored against any word the model knows.**

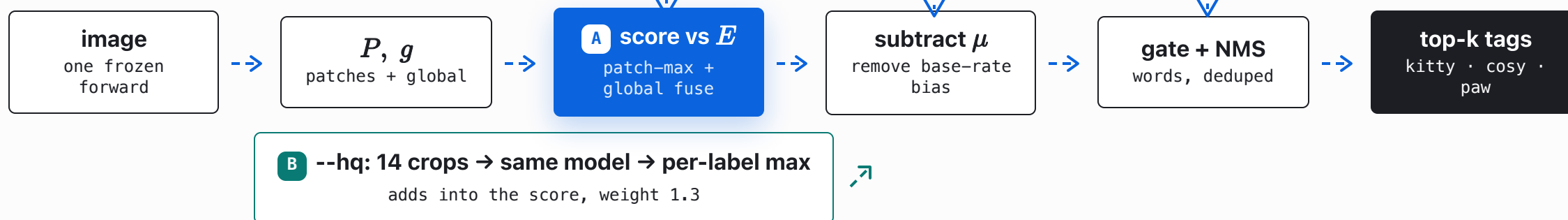
— THE PIPELINE

The complete tagger workflow as test-time compute.

OFFLINE



PER IMAGE



Every box is the frozen model or plain arithmetic: **A** reads more of one pass; **B** runs new passes on new pixels.

The label set is the tokenizer's own vocabulary.

No curated tag list. Take all **128,260** tokens and encode each one as text into a label matrix

$$E = [e_\ell]_{\ell \in V}, \quad e_\ell = \text{encode_text}(\ell) \in \mathbb{R}^{768}, \quad |V| = 128,260$$

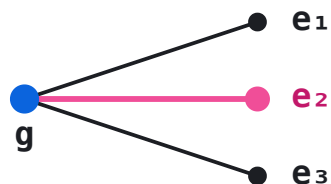
One matmul now scores any image against the entire vocabulary; fragments are filtered later by a word-start gate (step 4).

```
wrong: image_vec · embed_tokens.weight  
       → Tutor, avatar, PyTuple (garbage)  
right: image_vec · encode_text(token)  
       → kitty, cosy, plush, paw (aligned)
```

Why encode_text, not the embedding table? The model has `tie_word_embeddings = false`: the input-embedding table lives in a different space than the pooled output. You must encode each token as a *text string* to put labels in the image's space.

Score every patch.

GLOBAL POOLED

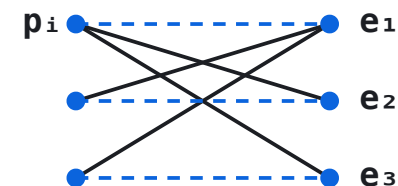


$$s_\ell = \langle g, e_\ell \rangle$$

one vector dominated by the most salient object

mAP 0.264

PER-LABEL MAX OVER PATCHES



$$s_\ell = 0.7 \max_{p \in P} \langle p, e_\ell \rangle + 0.3 \langle g, e_\ell \rangle$$

every label takes its max over all patches: late interaction (ColBERT), with patches as the token vectors

mAP 0.635

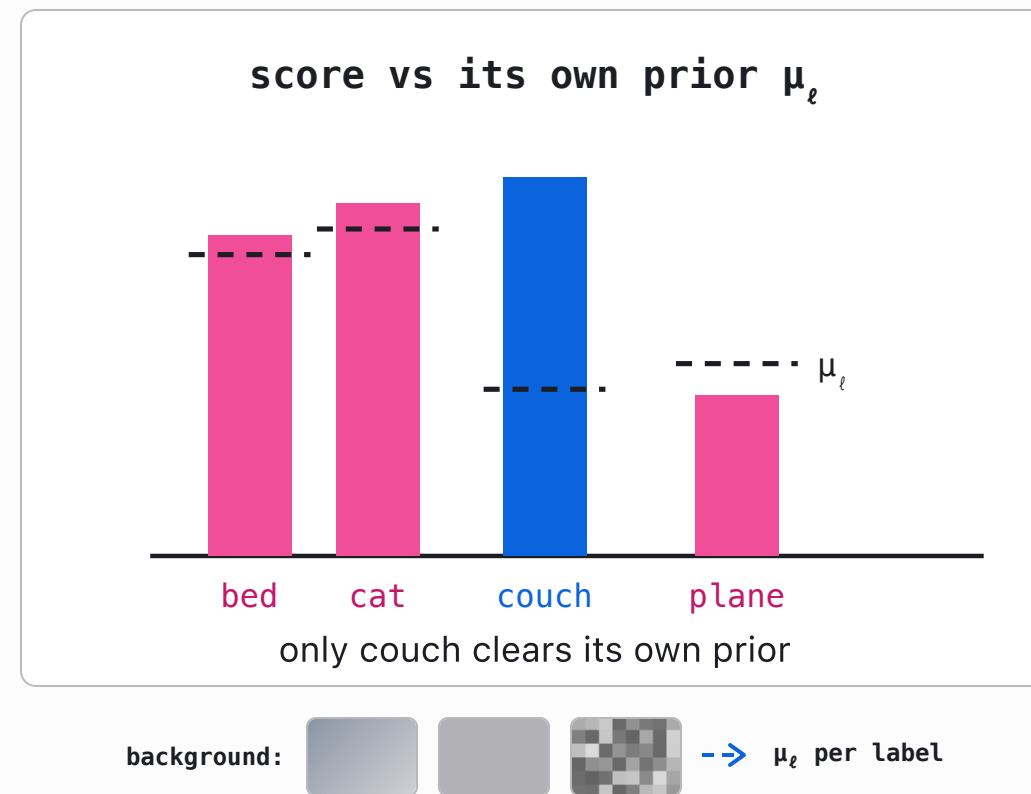
A PIAA's "[CLS] is not enough", confirmed: a small object fires on its own patch and survives the max. **+0.37 mAP**, from rows the pass already produced.

Subtract the per-label prior.

Raw cosine has a base-rate bias: generic words ("bed", "cat") score high on almost any image because they sit near the image modality. Estimate a per-label prior once from neutral background images, then subtract it:

$$\tilde{s}_\ell = s_\ell - \mu_\ell, \quad \mu_\ell = \mathbb{E}_{x \sim \text{background}} [s_\ell(x)]$$

No calibration set, no labels, one offline pass.



The tokenizer already knows what a word is.

Word-start gate. Byte-BPE tokenizers encode a leading space as the glyph `Ġ`; a token beginning with it starts a new word: `Ġcat` is a whole word, `GetComponent` is a fragment. That one built-in signal is the gate; no external dictionary.

Embedding-NMS. Synonyms and multilingual variants (`кот` / `Cat` / `кот` / `cat`) collapse into one, using cosine in the model's own space. Walk labels by score; keep ℓ only if $\max_{k \in \text{kept}} \langle e_\ell, e_k \rangle < 0.6$. Again: no lookup table, just the geometry.

raw vocabulary

128,260 tokens

`Ġkitty` · `Ġcat` · `GetComponent` · `_cpp` · `猫` · `quisites`

↓ word-start gate: keep tokens with the leading-space mark `Ġ`

whole words

25,465

`kitty` · `cat` · `kitten` · `chatte` · `kitt` · `cats` · `kittens` · `cosy` · ...

↓ embedding-NMS: walk down by score; drop any word with cosine ≥ 0.6 to one already kept

distinct concepts

top-k

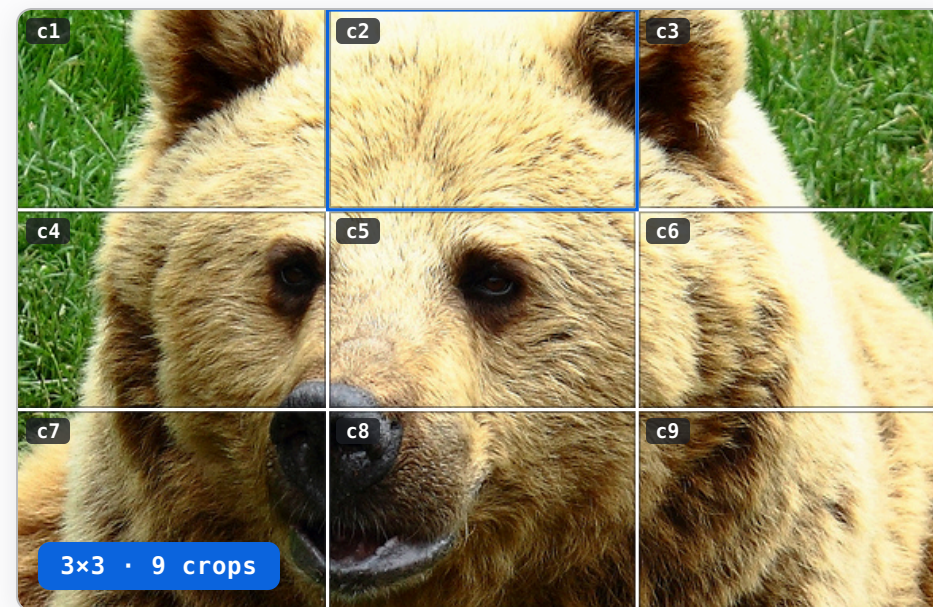
`kitty` · `cat` · `kitten` · `chatte` · `cats` · `cosy` · `plush` · `crib`

Multi-crop re-encoding.

Class-Wise Reranking (CWR), the crop-and-rescore idea from TagCLIP (AAAI'24), extended to a full-coverage grid of 14 crops (overlaid right) through the same frozen model. A crop is a *new input image*, itself re-patchified by the tower; a small object fills whichever crop contains it, so weak signal becomes strong.

$$S_\ell = \tilde{s}_\ell + 1.3 \left(\max_{c=1..14} s_{c,\ell} - \mu_\ell \right), \quad s_{c,\ell} = \max \left(\max_{p \in P_c} \langle p, e_\ell \rangle, \langle g_c, e_\ell \rangle \right)$$

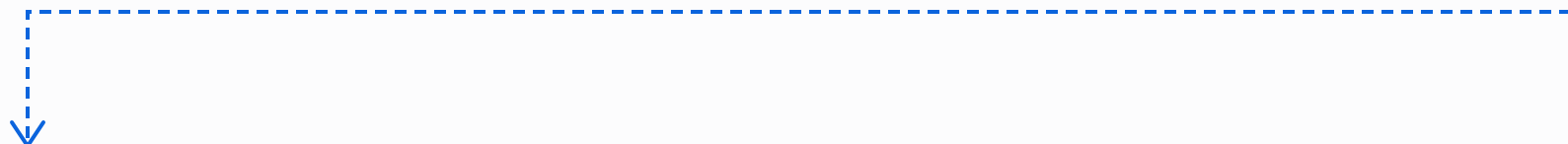
B Per-label **max**, not averaging: **averaging hurt**. The outlier crop is the evidence.



fast: **wolf**, **roar**, **muzzle** → --hq: **fur**, **bear**, **muzzle**
 multi-crop re-encoding corrects the single-pass misreading

The entire tagger is one scoring function.

the same score , applied to each crop as a fresh image


$$S(\ell) = \underbrace{0.7 \left(\max_{p \in P} \langle p, e_\ell \rangle - \mu_\ell \right)}_{\text{patch evidence}} + \underbrace{0.3 \left(\langle g, e_\ell \rangle - \mu_\ell^g \right)}_{\text{global context}} + \underbrace{1.3 \left(\max_{c=1..14} s_\ell(\text{crop}_c) - \mu_\ell \right)}_{\text{multi-crop re-encode}}$$

A Patch evidence: algebra over rows the forward pass already produced. Free. +0.37 mAP.

A Global context: the pooled vector, de-biased by its own background prior. Free.

B Multi-crop: 14 fresh forward passes on new pixels. 14× the cost. +0.075 mAP.

then $\ell \in \text{word-gated vocabulary} \rightarrow \text{embedding-NMS } (\tau = 0.6) \rightarrow \text{top-}k$

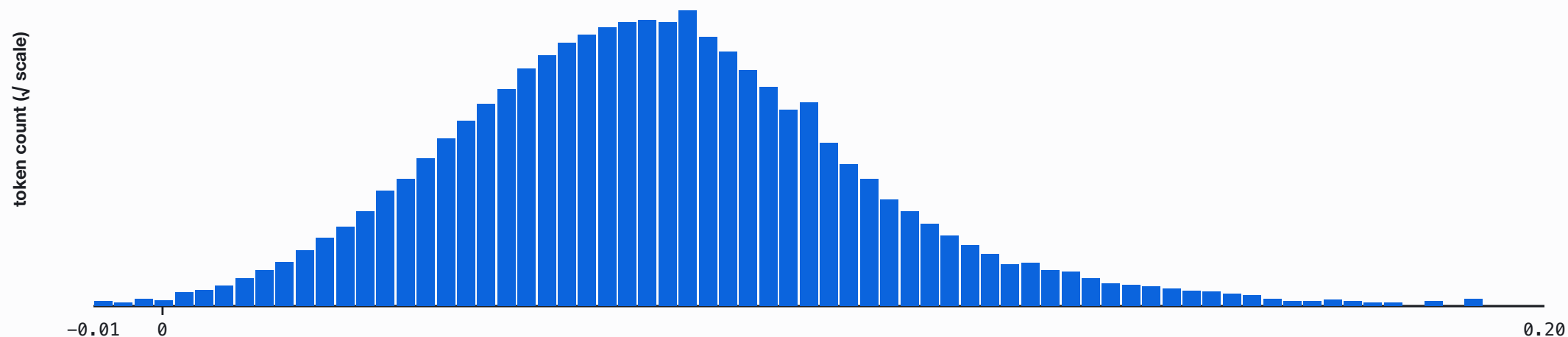
No head, no logits, no learned threshold: **three fixed weights, two background priors, and max operations over frozen-model outputs.**

— THE EQUATION, VISUALIZED

The pipeline sharpens a distribution.

1 · raw fused score

all 128,260 tokens · 0.7 patch-max + 0.3 global



a smooth bell: every token is somewhat close to every image

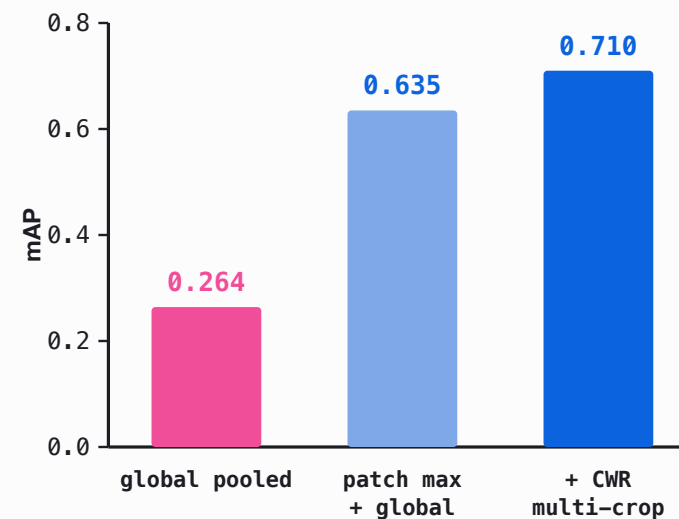
Real pipeline output, one image, all 128,260 tokens. Tagging is distribution sharpening.

RESULTS

Results on COCO-150.

METHOD	P@1	P@3	R@5	MAP
global pooled	0.433	0.289	0.449	0.264
patch max + global	0.753	0.427	0.631	0.635
softpool	0.773	0.449	0.649	0.608
+ CWR multi-crop	0.813	0.476	0.680	0.710

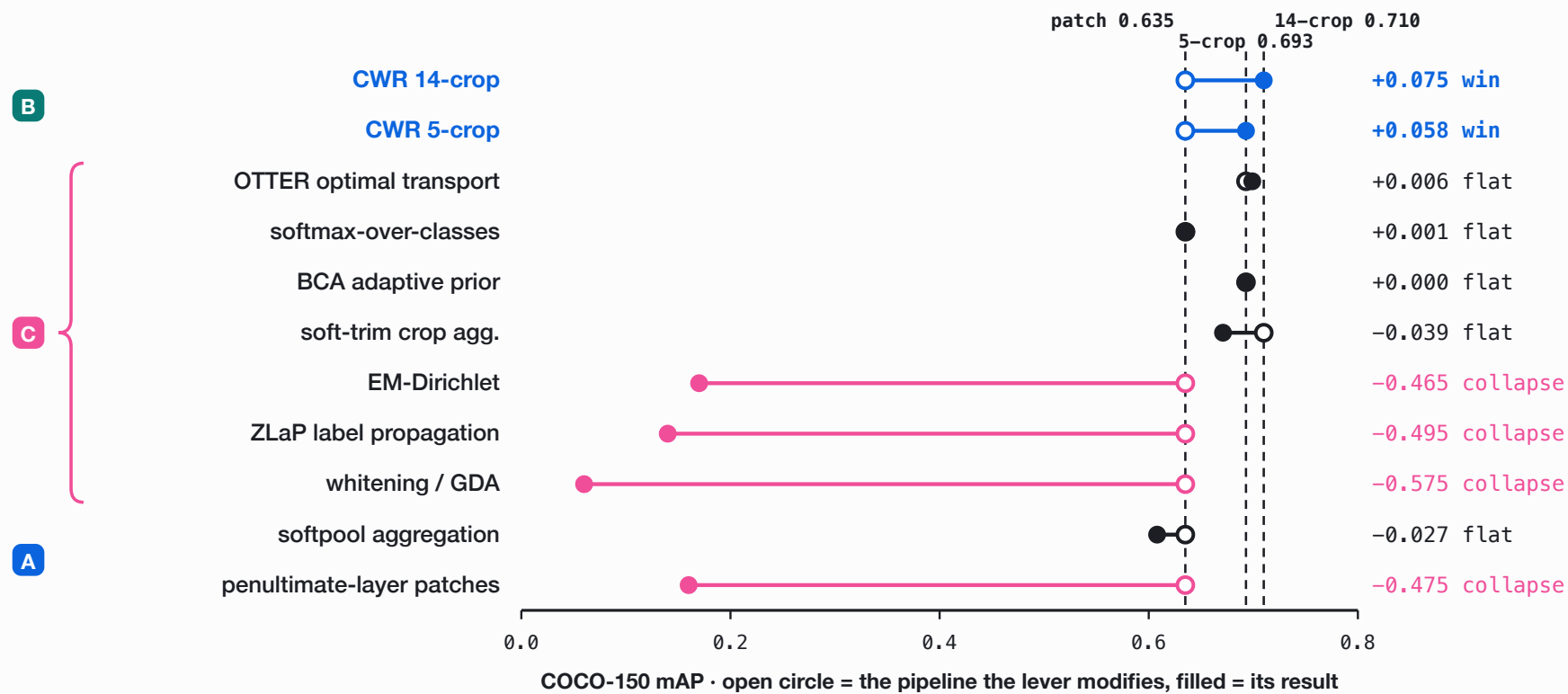
150 COCO-val images, real multi-label ground truth, avg 2.93 labels/image. softpool trades mAP for top-k precision; the ladder tracks the max-pool path.



mAP: global → patch → +CWR

Does bootstrapping scale accuracy at test time?

One example: **OTTER** re-balances scores toward a target class distribution — a pure re-arrangement. Its baseline (open circle) is the 5-crop result **0.693**, not 0.635; it lands at 0.699.



Only new information moves accuracy.

Bootstrap methods were designed for weakly calibrated, single-label, two-tower CLIP. This encoder's space is already aligned, calibrated, and multi-label: **there is nothing left to recover.**

BOOTSTRAP

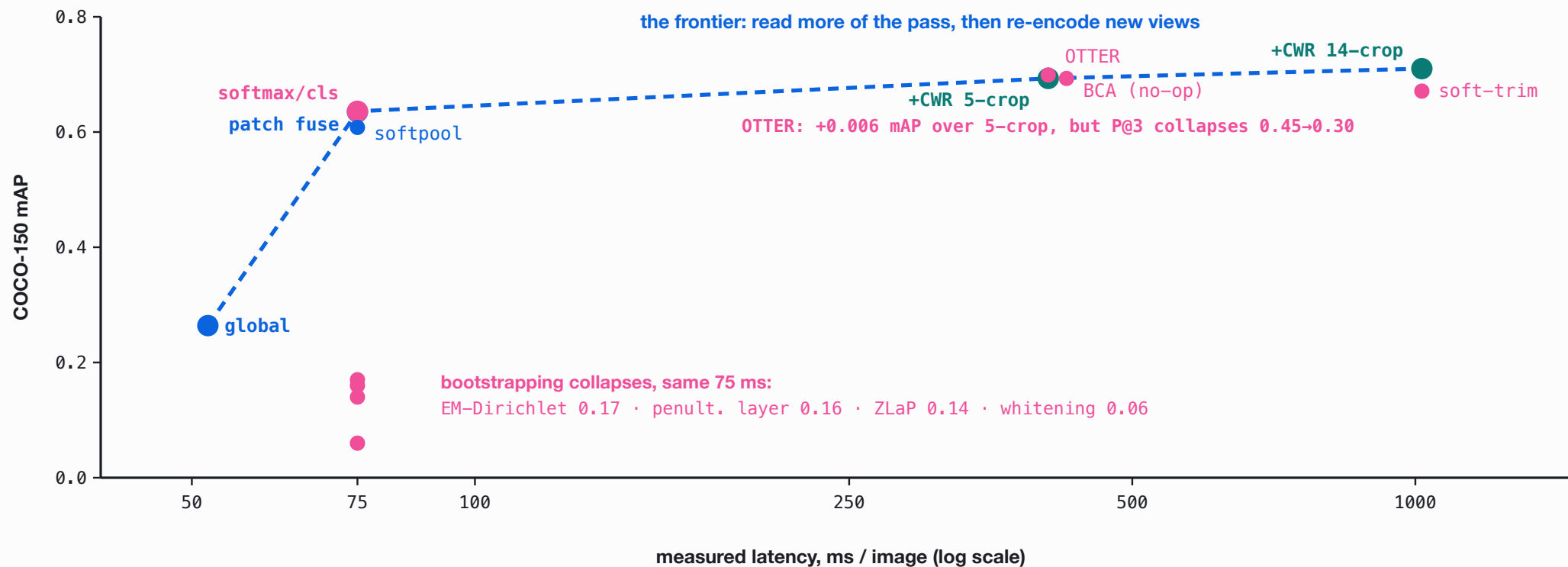
whitening 0.06 · ZLaP 0.14 · EM-Dirichlet 0.17 · OTTER
+0.006 over its own base. None improves the pipeline it
modifies.

FEEDING NEW INFORMATION

CWR crops → mAP 0.710, P@1 0.813. The only intervention
that improves results.

The ceiling is the 1B model itself. Real test-time compute here means
giving the model more to look at, not re-arranging what it already saw.

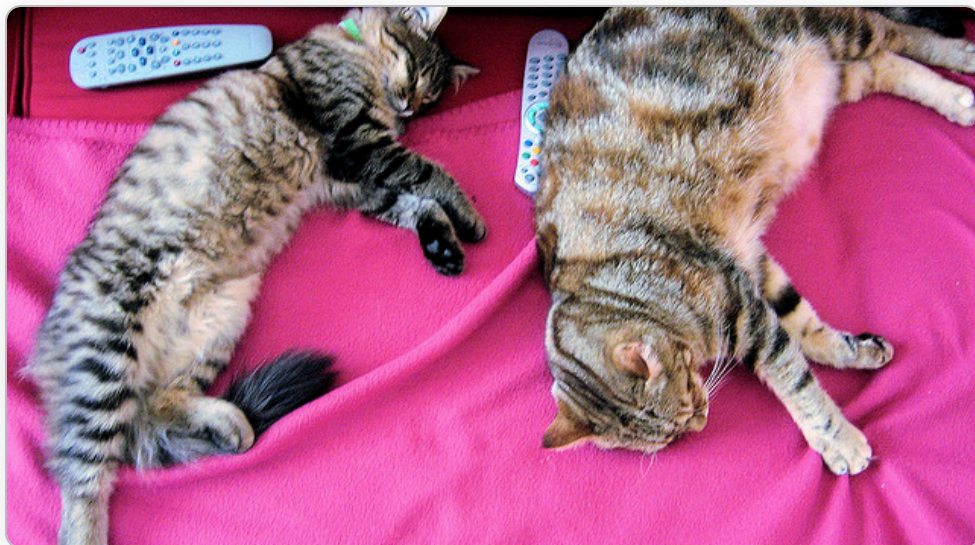
Accuracy versus test-time compute.



The frontier is traced by reading more of the pass and re-encoding new views;
bootstrapping sits at the same cost, at or below it (<0.1 ms/img, measured).

Extend to n-grams.

Every tag so far is a single word. Can more **test-time compute** turn it into a grounded **noun phrase** — no POS tagger, no grammar, no new model?

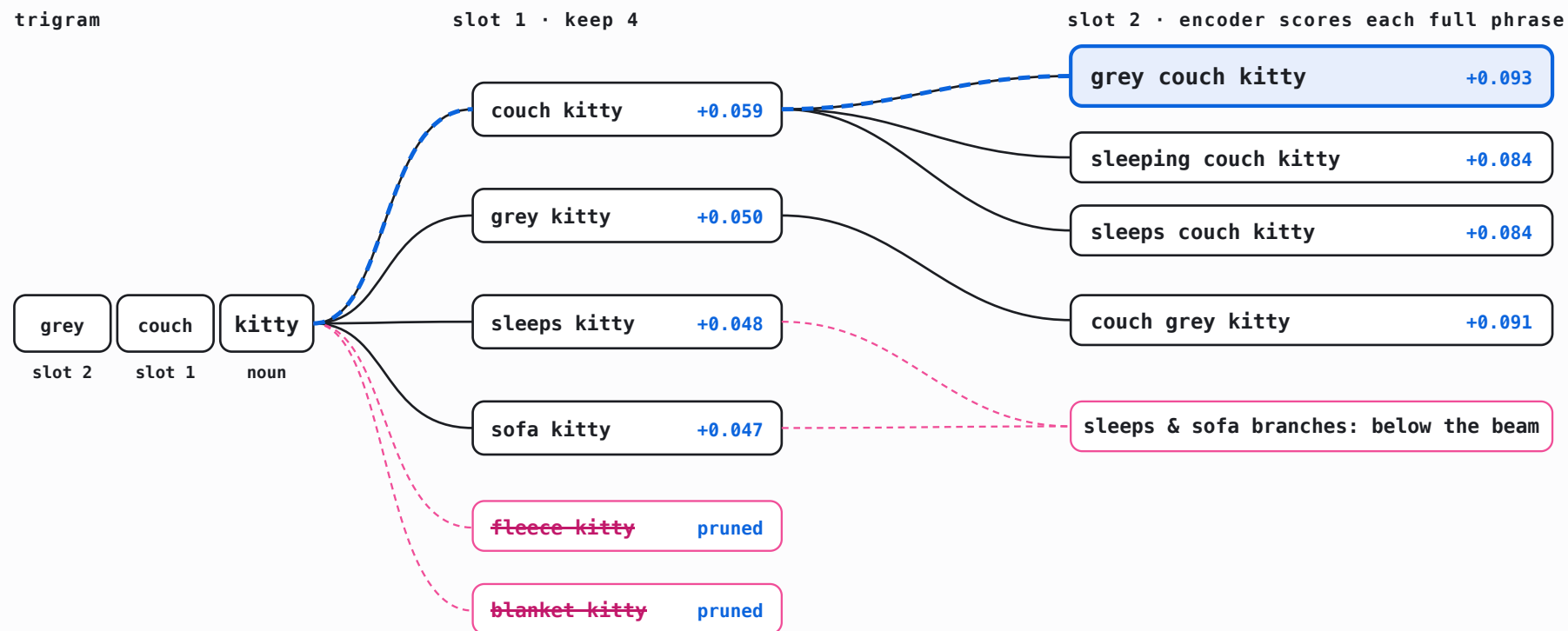


THE SAME FOUR TAGS, N = 1, 2, 3

N=1	kitty	cosy	plush	crib
N=2	couch kitty	black cosy	grey plush	grey crib
N=3	grey couch kitty	sleeping black cosy	grey sleeping plush	sleeps grey crib

Beam search over n-gram slots.

Each slot is filled one at a time. At every level the encoder scores the full phrase against the region $(\langle E(\text{phrase}), \text{region} \rangle - \langle E(\text{kitty}), \text{region} \rangle)$; the best 4 across *all* expansions survive. No backtracking — the winner is read off the final beam.



The winning path re-orders its own words: "grey couch kitty" beats "couch grey kitty". Word order, resolved by an embedding model with no grammar.

n-gram tagging in action.



DEFAULT onstage attendees livest concert presenter
CROPS onstage venue ceiling projector theater
N=2 conference onstage projector attendees attendees livest projector concert projector presenter
N=3 conference crowd onstage lobby projector attendees projector attendees livest projector lobby concert attendees projector presenter



DEFAULT driv rims ford traction sidew
CROPS carro rims windshield steering beach
N=2 windshield driv windshield rims sail ford windshield traction blue sidew
N=3 blackjack windshield driv dealership windshield rims windshield yacht ford yacht sedan traction blue car sidew



DEFAULT attendees passengers protestors migrants crowded
CROPS attendees commuters demonstrators synagogue crowded
N=2 protester attendees demonstrators passengers passengers protestors attendees migrants demonstrators crowded
N=3 passengers protester attendees crowded demonstrators passengers passenger attendees protestors passengers protester migrants
passenger demonstrators crowded

Default, crop re-encoding, beam-searched n-grams. Top-5 per mode, unstaged photos. [Verbatim, unfiltered.](#)

Versus the recent tagging literature.

LINE OF WORK	THEIR SETUP	OUR DELTA
Tag2Text / RAM / RAM++	trained tagging models, curated ~6.4k tag list	training-free, label space = tokenizer vocab
TagCLIP	CLIP two-tower, penultimate layer, DMAR+CWR	single omni model, last layer is the output space
PIAA	patch-level scoring ("[CLS] is not enough") + GDA whitening	patch thesis confirmed (+0.37 mAP); whitening collapses (0.06)
recent calibration (OTTER, BCA)	calibration set or target prior to de-bias	one offline background prior, zero calibration

RQ1: a frozen embedding model does become a tagger, training-free. **RQ2:** only compute that adds information scales it. Two questions, answered.

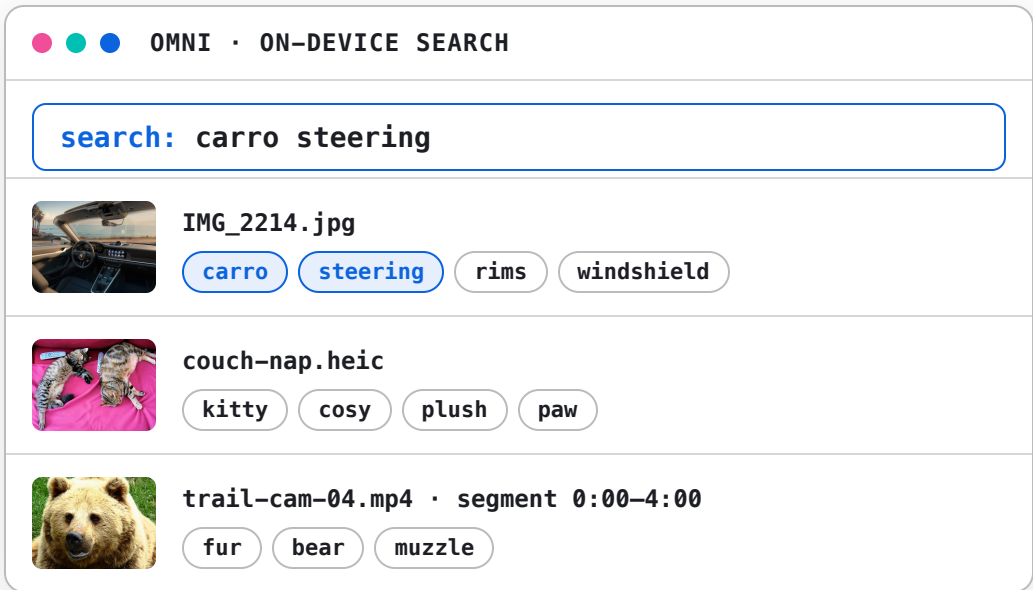


DEPLOYMENT

On-device deployment in Omni.

The Python study became a feature of **Omni**, a native on-device semantic-search app (Swift + MLX, same frozen model). The port reuses the architecture directly:

- | SAME FORWARD PASS the patch rows already exist for the file's *embedding*; tagging adds one matmul.
- | ~0.6 MS / IMAGE about 4% overhead on the embed step. Tagging is effectively free at index time.
- | TAGS = SNIPPETS tags land in the search snippet: media becomes *keyword*-searchable.
- | SELF-CALIBRATING the background prior μ is estimated on-device from the first 64 images it sees, then frozen.



tag chips as they appear in results; the third row is a video segment

Images, video, and scanned documents.



IMAGE

Tagged at **index time**, inside the embedding forward. Patch-max + global, prior-centered, gated, NMS-deduped.



HQ IMAGE

A background **re-tag pass** adds the production CWR variant: 5 crops, per-label max.



VIDEO

32 frames per **240-second segment**, one sequence through the tower: patch-max pools over **space and time**.



SCANNED PDF

Pages render to images, same path: **per-page tags** ("invoice, table, signature") as the snippet.

A capability the model was never trained for, realized at inference.

Two talks, one thesis. A frozen encoder holds more capability than its training objective exposes, and you unlock it with **test-time compute**, not more parameters.

RETRIEVAL TALK

recombine the geometry for more **relevance** on the task it was trained for.

THIS TALK

recombine the geometry for an **emergent new task** it was never trained for: tagging.

Test-time compute scales only when it **supplies the model with new information**. Bootstrapping an already-good representation yields nothing.

Scaling test-time compute of an embedding model **unlocks tasks beyond retrieval.**

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GITHUB REPO

